

Spring Boot Assignment Report

Overview

This application manages two entities: **Books** and **Authors**. It supports reading all books, adding new books, and updating existing book details, with a user interface implemented in JSP.

1. Entity Relationship Design

- **Entities:** Author and Book
- **Relationship:** OneToMany from Author to Book (an author can have multiple books), and ManyToOne from Book to Author.
- **Primary Keys:** id auto-generated using identity strategies.
- **Constraints:** Book isbn and Author name are marked as unique, and properties like title, name, and email are non-nullable.

```
@Entity
@Table(name = "authors")
public class Author {
    @Id @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    // ...
    @OneToMany(mappedBy = "author", cascade = CascadeType.ALL, fetch =
FetchType.LAZY)
    private List<Book> books;
}

@Entity
@Table(name = "books")
public class Book {
    @Id @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    // ...
    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "author_id", nullable = false)
    private Author author;
}
```

2. Implementation Details

Database Initialization

The `DataLoader` class implements `CommandLineRunner` to populate the H2 database with 10 sample authors and 10 sample books upon application startup.

Read Operation

- **Controller:** `@GetMapping("/books")` fetches the list of books via `libraryService.getAllBooksWithAuthor()` and maps it to `book-list.jsp`.

- **Repository Custom Query:** To avoid the N+1 problem, an inner join is explicitly requested and used:

```
@Query("SELECT b FROM Book b INNER JOIN FETCH b.author")
List<Book> findAllBooksWithAuthor();
```

- **View:** `book-list.jsp` displays the entities in an HTML table formatted with CSS.

Book List

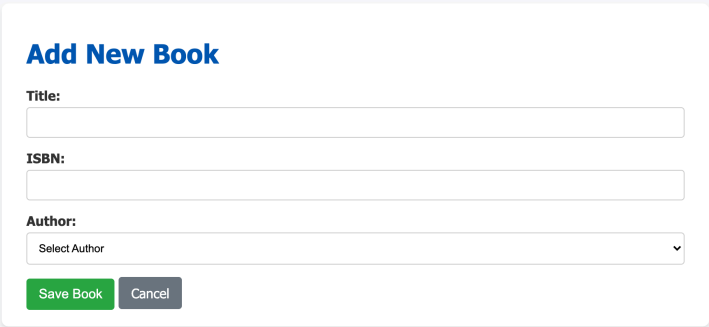
Add New Book

ID	Title	ISBN	Author	Actions
1	Book Title 1	ISBN-1001	Author 1	Edit
2	Book Title 2	ISBN-1002	Author 2	Edit
3	Book Title 3	ISBN-1003	Author 3	Edit
4	Book Title 4	ISBN-1004	Author 4	Edit
5	Book Title 5	ISBN-1005	Author 5	Edit
6	Book Title 6	ISBN-1006	Author 6	Edit
7	Book Title 7	ISBN-1007	Author 7	Edit
8	Book Title 8	ISBN-1008	Author 8	Edit
9	Book Title 9	ISBN-1009	Author 9	Edit
10	Book Title 10	ISBN-10010	Author 10	Edit

Create Operation

- **Controller:** `@GetMapping("/books/add")` provides the view `book-form.jsp`.
`@PostMapping("/books/add")` saves the `Book` linked to a selected `Author`.
- **Exception Handling:** Integrity constraint violations (e.g., duplicate ISBN) are caught and handled by redirecting the user back to the form with a descriptive error message using

```
RedirectAttributes.addFlashAttribute() .
```



The image shows a web form titled "Add New Book" in blue text. Below the title are three input fields: "Title:" with an empty text box, "ISBN:" with an empty text box, and "Author:" with a dropdown menu showing "Select Author". At the bottom of the form are two buttons: a green "Save Book" button and a grey "Cancel" button.

Update Operation

- **Controller:** `@GetMapping("/books/edit/{id}")` populates a form with the book details.
`@PostMapping("/books/edit/{id}")` processes the submitted edits.
- **Exception Handling:** Similar to creation, handled smoothly in the controller for uniqueness checks.



The image shows a web form titled "Edit Book" in blue text. Below the title are three input fields: "Title:" with a text box containing "Book Title 1", "ISBN:" with a text box containing "ISBN-1001", and "Author:" with a dropdown menu showing "Author 1". At the bottom of the form are two buttons: a green "Update Book" button and a grey "Cancel" button.

3. Testing

Comprehensive test suites are written leveraging JUnit and Mockito:

- **Repository Tests:** Validates the custom `JOIN` query using `@DataJpaTest` .
- **Service Tests:** Mocks the repository dependencies using `@InjectMocks` and `@Mock` to test `getAllAuthors()` , `getAllBooksWithAuthor()` , `saveBook()` , and `updateBook()` . All tests run successfully without any errors.

4. Challenges Faced & Solutions

1. Handling N+1 Select Problems in the View:

- *Challenge:* Displaying the author's name next to the book's title naively triggered a new query per book.
- *Solution:* Leveraged the JPQL custom query `SELECT b FROM Book b INNER JOIN FETCH b.author` within the repository to eagerly load relations in a single optimized query.

2. JSP Rendering in Spring Boot 3:

- *Challenge:* Spring Boot primarily suggests Thymeleaf, making standard JSP setup slightly tricky due to modern embedded tomcat structures.
- *Solution:* Successfully managed by properly specifying `<packaging>jar</packaging>` and importing Tomcat Embed Jasper and standard Jakarta JSTL dependencies.

3. Database Constraints Handling:

- *Challenge:* Providing a user-friendly error to duplicate entries (like ISBN).
- *Solution:* Wrapped service calls inside a try-catch for `DataIntegrityViolationException` in the Controller layer, keeping the application stable.

5. Github URL

- <https://github.com/Suhassk205/Springboot-assignment-bits>